Lillian Barrett Hughes Wyatt

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Research Vision

Creating quantum defects and heterostructures that enable robust coherence and controlled interactions by integrating precision synthesis, characterization, and fabrication with quantum measurements.

Professional Experience

- **California Institute of Technology** — *Nov. 2025 - Present*
  AWS Quantum Postdoctoral Scholar — Pasadena, CA, U.S.A.

Education

- **University of California Santa Barbara** — *Sep. 2019 - Sep. 2025*
  Ph.D. in Materials — Santa Barbara, CA, U.S.A.
  Advisors: Professor Ania C. Bleszynski Jayich and Professor Kunal Mukherjee (Stanford)
- **University of Richmond** — *Aug. 2015 - May 2019*
  B.S. in Interdisciplinary Physics, concentration in Chemistry — Richmond, VA, U.S.A.
  Summa cum laude
  Advisor: Professor Michael C. Leopold

Honors and Awards

- **Materials Research Society Graduate Student Silver Award,** MRS Spring Meeting (2025)
- **Best Student Oral Presentation Award,** MRS Fall Meeting, Diamond Symposium (2023)
- **First Place Student Presentation Award,** New Diamond and Nano Carbon Conference (2023)
- **Best Student Oral Presentation Award,** European MRS Meeting, Diamond Symposium (2023)
- **UCSB Materials Department Service Award,** Bright Horizon Global Foundation (2022)
- **NSF Graduate Research Fellowship** (2021)
- **NDSEG Fellowship** (2021)
- **UCSB NSF Quantum Foundry Graduate Fellowship** (2021)
- **NSF GRFP Honorable Mention** (2019)
- **UC Regents Fellowship,** Materials Department, UCSB (2019)
- **David C. Evans Award for Outstanding Achievement in Scholarship,** College of Arts and Sciences, UR (2019)
- **Robert E. Loving Award for Top Graduating Student,** Department of Physics, UR (2019)
- **Goldwater Scholarship** (2018)
- **Phi Beta Kappa,** UR (2018)
- **Multiple departmental awards in Physics and Chemistry,** UR (2017–2018)

Publications

First-author:

[1] **L. B. Hughes Wyatt**,* S. Parthasarathy,* I. Kantor,* C. K. Kim, L. Chen, T. A. Morrison, J. Ahlers, K. Mukherjee, A. C. Bleszynski Jayich. Creation of depth-confined, shallow nitrogen-vacancy centers in diamond with tunable density. *Appl. Phys. Lett.* **128**, 204006 (2026). doi.org/10.1063/5.0316283

[2] **L. B. Hughes**, S. A. Meynell, W. Wu, S. Parthasarathy, L. Chen, Z. Zhang, Z. Wang, E. J. Davis, K. Mukherjee, N. Y. Yao, and A. C. Bleszynski Jayich. A Strongly Interacting, Two-Dimensional, Dipolar Spin Ensemble in (111)-Oriented Diamond. *Phys. Rev. X* **15**, 021035 (2025). doi.org/10.1103/PhysRevX.15.021035

[3] **L. B. Hughes**, Z. Zhang, C. Jin, S. A. Meynell, B. Ye, W. Wu, Z. Wang, E. J. Davis, T. E. Mates, N. Y. Yao, K. Mukherjee, and A. C. Bleszynski Jayich. Two-dimensional spin systems in PECVD-grown diamond with tunable density and long coherence for enhanced quantum sensing and simulation. *APL Materials* **11**, 021101 (2023). doi.org/10.1063/5.0133501
***Selected as a featured article.**

[4] **L. B. Hughes**,* N. Labban,* G. Conway, J. Pollock, and M.C. Leopold. Adaptable Xerogel-Layered Amperometric Biosensor Platforms on Wire Electrodes for Clinically Relevant Measurements. *Sensors* **19**, 2584 (2019). doi.org/10.3390/s19112584

Co-authored:

[5] E. Postelnicu, **L. B. Hughes Wyatt**, T. Ngyuen, S. A. Meynell, C. Jilly, P. Wallace, A. Barnum, A. C. Bleszynski Jayich, K. Mukherjee. Harnessing diamond surface features for dense and aligned NV ensembles. *Nano Lett.* (2026). doi.org/10.1021/acs.nanolett.6c02155

[6] C. C Newsom, **L. B Hughes Wyatt**, B. L Green, A. C Bleszynski Jayich, M. E Newton. 200 keV energy electron irradiation of single crystal diamond: Quantification of vacancy and nitrogen-vacancy production. *Phys. Rev. B.* **114**, 065203 (2026). doi.org/10.1103/hls1-92jc

[7] H. Gao, N. T. Leitao, S. Dandavate, **L. B. Hughes Wyatt**, P. Put, M. Mammen, L. S. Martin, H. Park, A. C. Bleszynski Jayich, M. D. Lukin. Dressed-state Hamiltonian engineering in a strongly interacting solid-state spin ensemble. *Phys. Rev. Lett.* **136**, 200802 (2026). doi.org/10.1103/d1kx-hy93

[8] H. Lee,* H. C. Kleidermacher,* A. J. M. Stein,* H. Oh, **L. B. Hughes Wyatt**, C. K. Kim, L. Basso, A. Mounce, S. S. Su, M. Titze, A. C. Bleszynski Jayich, J. Vuckovic. Quantum nanophotonic interface for tin-vacancy centers in thin-film diamond. *Phys. Rev. Appl.* **25**, 044074 (2026). doi.org/10.1103/vwt7-b64x

[9] H. Oh,* V. Dharod,* C. Padgett,* **L. B. Hughes Wyatt**, J. Venkatraman, S. Parthasarathy, E. Osipova, I. Hedgepeth, J. V. Cady, L. Basso, Y. Wang, M. Titze, E. S. Bielejec, A. M. Mounce, D. Bouwmeester, A. C. Bleszynski Jayich. Spin-embedded diamond optomechanical resonator with a mechanical quality factor exceeding one million. *Optica* **13**, 485-490 (2026). doi.org/10.1364/OPTICA.577541

[10] B. Edelman, C. Sheppard, L. Chuidian, A. Vinnikov, F. Bevc, **L. B. Hughes**, K. Kittredge, C. Parish, M. Leopold. Design of Nanomaterial-Based Sensors for Enhanced Halogen Bonding. *ACS Omega* **11**, 4006-4020 (2026). doi.org/10.1021/acsomega.5c07542

[11] S. Kim, P. London, D. Yang, **L. B. Hughes**, J. Ahlers, S. Meynell, W. J. Mitchell, K. Mukherjee, A. C. Bleszynski Jayich. Scalable nanoscale positioning of highly coherent color centers in prefabricated diamond nanostructures. *Nat. Comm.* **16**, 9803 (2025). doi.org/10.1038/s41467-025-64758-4

[12] H. Gao, L. S. Martin, **L. B. Hughes**, N. T. Leitao, P. Put, B. Ye, H. Zhou, W. Wu, N. U. Koyluoglu1, S. Meynell, E. Davis, N. Yao, A. C. Bleszynski Jayich, H. Park, M. D. Lukin. Signal amplification in a solid-state quantum sensor via asymmetric time-reversal of many-body dynamics. *Nature* **646**, 68-74 (2025). doi.org/10.1038/s41586-025-09452-7

[13] W. Wu, E. J. Davis, **L. B. Hughes**, B. Ye, Z. Wang, D. Kufel, T. Ono, S. A. Meynell, M. Block. C. Liu, H. Yang, A. C. Bleszynski Jayich, N. Y. Yao. Spin squeezing in an ensemble of nitrogen-vacancy centers in diamond. *Nature* **646**, 74-80 (2025). doi.org/10.1038/s41586-025-09524-8

[14] S. Parthasarathy, M. Joos, **L. B. Hughes**, S. A. Meynell, T. Morrison, K. Mukherjee, D. Weld, A. C. Bleszynski Jayich. Role of oxygen in laser induced contamination of diamond-vacuum interfaces. *Phys. Rev. Appl.* **22**, 024067 (2024). doi.org/10.1103/PhysRevApplied.22.024067

[15] P. D. Reddy, L. Nordin, **L. B. Hughes**, A. K. Priedl, K. Mukherjee. Expanded stability of layered SnSe-PbSe alloys and evidence of displacive phase transformation from rocksalt in heteroepitaxial thin films. *ACS Nano* **18**, 13437-13449 (2024). doi.org/10.1021/acsnano.4c04128

[16] L. V. H. Rogers, S. T. Nguyen, J. H. Cox, K. Zervas, Z.Yuan, S. Sangtawesin, A. Stacey, C. Jaye, C. Weiland, A. Pershin, A. Gali, L. Thomsen, S. A. Meynell, **L. B. Hughes**, A. C. Bleszynski Jayich, X. Gui, R. J Cava, R. R. Knowles, N. P. de Leon. Diamond Surface Functionalization via visible light-driven CH activation for nanoscale quantum sensing. *PNAS* **121**, e2316032121 (2024). doi.org/10.1073/pnas.2316032121

[17] L. V. H. Rogers, **L. B. Hughes**, M. Xie, S. W. Kolkowitz, P. C. Maurer, A. C. Bleszynski Jayich, N. P. de Leon. Materials challenges for quantum technologies based on color centers in diamond. *MRS Bulletin* **46**, 623-633 (2021). doi.org/10.1557/s43577-021-00137-w
***Featured in a special collection of journal's most impactful publications.**

[18] S. A. Meynell, C. A. McLellan, **L. B. Hughes**, W. Wang, K. Mukherjee, and A. C. Bleszynski Jayich. Engineering quantum-coherent defects: the role of substrate miscut in chemical vapor deposition diamond growth. *Appl. Phys. Lett.* **117**, 194001 (2020). doi.org/10.1063/5.0029715
***Editor's pick.**

[19] A. K. Jaini, **L. B. Hughes**, M. Kitimet, K. J. Ulep, M. C. Leopold, and C. A. Parish. Halogen Bonding Interactions for Aromatic and Non-Aromatic Explosive Detection. *ACS Sensors* **4**, 389-397 (2019). doi.org/10.1021/acssensors.8b01246

In press, under review, and submitted:

[20] P. Put,* N. T. Leitao,* H. Gao,* C. Spaegele,* O. Makarova, **L. B. Hughes**, A. C. Maccabe, M. Mammen, B. Machielse, H. Zhou, S. Pustelny, A. C. Bleszynski Jayich, F. Capasso, L. S. Martin, H. Park, M. D. Lukin. Collective many-body dynamics in a solid-state quantum sensor controlled through nanoscale magnetic gradients. *In press at PRX.* (2025) arXiv:2506.11920

[21] Z. Zhang, T. Morrison, **L. B. Hughes**, W. Wu, R. Liu, D. Bluvstein, N. Yao, D. Fygenson, A. C. Bleszynski Jayich. Patterning programmable spin arrays on DNA origami for quantum technologies. *Submitted.* (2025) arXiv:2509.10760

[22] H. Gao, P. Put, N. T. Leitao, N. U. Köylüoğlu, A. Maccabe, M. Mammen, S. Dandavate, **L. B. Hughes Wyatt**, L. S. Martin, A. C. Bleszynski Jayich, H. Park, M. D. Lukin, Nanoscale magnetometry via collective many-body dynamics in diamond. *Submitted.* (2026) arXiv:2609.03039v1

[23] T. Ono,* W. Wu,* H. Yang, **L. B. Hughes Wyatt**, B. Brenner, C. Liu, C. Fan, C. R. Laumann, J. N. Hallén, E. J. Davis, A. C. Bleszynski Jayich, N. Y. Yao. Direct Observation of Dipolar-Driven Anisotropic Quantum Projection Noise in a Solid-State Spin Ensemble. *Submitted.* (2026) arXiv:2609.16106

[24] L. Chen,* S. Parthasarathy,* S. A. Meynell,* **L. B. Hughes Wyatt**, E. Postelnicu, H. Yang, Z. Wang, W. Wu, W. V. Peloso, C. K. Kim, C. R. Laumann, K. Mukherjee, N. Y. Yao, A. C. Bleszynski Jayich. Engineering and Probing a One-Dimensional Dipolar Spin Ensemble in Diamond. *Submitted.* (2026) arXiv:2609.19496

In preparation:

[25] **L. B. Hughes Wyatt**, A. A. Hossain, Y. Yan, A. J. Minnich. Atomic layer etching of niobium via alternating O_2 and H_2/SF_6 plasma exposure. *In preparation.*

[26] **L. B. Hughes Wyatt** and A. J. Minnich. Atomic layer etching of tantalum via cryogenic BCl_3 and Ar plasma exposure. *In preparation.*

[27] **L. B. Hughes Wyatt**, I. Chen, and A. J. Minnich. Atomic layer etching of electronic and quantum materials. *In preparation.*

Patents

[1] **L. B. Hughes**, A. C. Jayich, S. Meynell, N. Yao, K. Mukherjee, W. Wu, E. J. Davis, "Strongly Interacting, Two-Dimensional, Dipolar Spin Ensemble in 111-Oriented Diamond for Quantum Sensing and Simulation," U.S. Patent Application Serial No. 19/647,295. Filed on April 14, 2026.

Invited Presentations

- **Materials Research Society Spring Meeting,** "Engineering interacting spin systems in diamond for quantum technologies," Apr. 2026, Honolulu, HI, U.S.A.
- **American Physical Society Global Physics Summit,** "Engineering interacting spin systems in diamond for quantum technologies," Mar. 2026, Denver, CO, U.S.A.
- **Future Leaders Seminar Series, Northwestern University Materials Department,** "Creating coherent NV centers in diamond for quantum technologies," Apr. 2024, Virtual.
- **Hasselt Diamond Workshop - SBDD XXVIII,** "Creating controlled NV systems in (100) and (111) diamond for quantum technologies," Feb. 2024, Hasselt, Belgium.
- **European Materials Research Society Meeting,** "Creating coherent spin systems in PECVD diamond for enhanced quantum sensing and simulation," Jun. 2023, Strasbourg, France.

Contributed Presentations

- **Materials Research Society Fall Meeting,** "Atomic Layer Processing for Superconducting Electronic Devices," Dec. 2026, Boston, MA, U.S.A.
- **Materials Research Society Spring Meeting,** "Creating Strongly Interacting Dipolar Spin Ensembles in Diamond for Quantum Technologies," Apr. 2025, Seattle, WA, U.S.A.
- **Materials Research Society Fall Meeting,** "Two-dimensional spin systems in PECVD-grown diamond with tunable density and long coherence for enhanced quantum sensing and simulation," Nov. 2023, Boston, MA, U.S.A.
- **Electronic Materials Conference,** "Two-Dimensional Spin Systems in PECVD-Grown Diamond with Tunable Density and Long Coherence for Enhanced Quantum Sensing and Simulation," Jun. 2023, Santa Barbara, CA, U.S.A.
- **New Diamond and Nano Carbon Conference,** "Two-dimensional spin systems in PECVD-grown diamond with tunable density and long coherence for enhanced quantum sensing and simulation," Jun. 2023, East Lansing, MI, U.S.A.
- **Hasselt Diamond Workshop - SBDD XXVI,** "Two-dimensional spin systems in CVD diamond for enhanced quantum sensing and simulation," Feb. 2022, Hasselt, Belgium.
- **Electronic Materials Conference,** "Understanding the Inhomogeneous Incorporation of Nitrogen in δ-Doped CVD Diamond Using Spatially Resolved Microanalysis," Jun. 2020, Virtual.

Teaching Interests and Experience

- **Teaching Interests**
  - **Materials Science and Engineering**: materials science, electronic properties of materials, thermodynamics, solid-state materials, quantum materials, thin-film materials, defects in solids, crystallography and diffraction
  - **Physics**: introductory physics, quantum mechanics, electricity and magnetism, solid-state physics, statistical mechanics, physics of matter/intro condensed matter

- **Chemistry**: general chemistry, physical chemistry, quantum mechanics, materials science, nanomaterials, quantum materials, inorganic chemistry
- **Electrical and Computer Engineering**: micro and nanofabrication, quantum computing and information, semiconductor devices, engineering quantum mechanics

- **Teaching Assistant: Introduction to Quantum Mechanics (MATRL 200Q), UCSB Materials Department** *Fall 2020*
  Teaching Assistant for Professor Anton Van der Ven
  - Graded homework, exams, and prepared weekly recitations for students to review course material and discuss problem sets.
- **Reader: Engineering Quantum Mechanics (MATRL 211B), UCSB Materials Department** *Spring 2022 and Spring 2024*
  Reader for Professor Chris Van de Walle
  - Graded homework, exams, and prepared weekly recitations for students to review course material and discuss problem sets.
  - Gave a substitute lecture on time-dependent perturbation theory.
- **Teaching Assistant: Cleanroom Training Program** *Aug. 2023 - Jan. 2024*
  Central Coast Partnership for Regional Industry-focused Micro/Nanotechnology Education, UCSB [🌐]
  - Led cohorts of students in two 40-hour programs focused on wafer fabrication processes in the cleanroom.
  - Taught hands-on skills including photolithography, metal deposition, dry and wet etching, and device inspection/characterization.
  - Served primarily URM and community college students.

Undergraduate Mentoring

- **Isaac Kantor,** UCSB (2024-2025) *Awarded the Goldwater Scholarship (2026).*
- **Ekaterina Osipova,** UCSB (2021-2025)
- **Ian McGuire,** Quantum Foundry REU (Summer 2024) *Now Ph.D. student in Physics at Brown University.*
- **Melika Dabiri,** Quantum Foundry REU (Summer 2024)
- **Haopu Yang,** UCSB (2022-2023) *Now Ph.D. student in Physics at Harvard University.*

Professional Service

- **Editorial and Peer Review**
  Reviewer for Journal of Applied Physics, Journal of Materials Chemistry C, Advanced Quantum Technologies, Frontiers in Quantum Science and Technology, Applied Research.
- **President, Treasurer, and Committee Member - Materials Scientist Association (MSA)** *Jan. 2021 - Sep. 2023*
  Materials Department, UCSB [🌐]
  - Led student organization, planned prospective graduate student recruitment events, interfaced with department leadership.
  - Goals of promoting community and professional development for students in the Materials department.
  - Wrote a proposal and received grant funding for mental health-focused programming for students in the department.
- **Outreach Programs** *Jan. 2021 - Sep. 2024*
  Quantum Foundry, Materials Research Lab, and Center for Science and Engineering Partnerships, UCSB [🌐]
  - Directed outreach project building Quantum Kits for middle school teachers. Led a team of 10 peers to develop lessons and demos.
  - Premiered the developed activities at World Quantum Day and in local middle schools.
  - Participated in volunteering and outreach programs at local elementary schools.
- **Writing Consultant** *Sep. 2021 - Nov. 2022*
  Graduate Writing Center, UCSB
  - Reviewed application essays through individual consultations with students applying for the NSF GRFP and NDSEG fellowships.
  - Provided tailored feedback as well as general strategies for successful proposal writing.

References

1. **Ania C. Bleszynski Jayich**
   Professor, Department of Physics, University of California Santa Barbara
   Relationship: Ph.D. Thesis Advisor
   Email: ania@physics.ucsb.edu
2. **Kunal Mukherjee**
   Assistant Professor, Department of Materials Science and Engineering, Stanford University
   Relationship: Ph.D. Thesis Co-Advisor
   Email: kunalm@stanford.edu
3. **Austin J. Minnich**
   Professor, Department of Mechanical Engineering and Applied Physics, California Institute of Technology
   Relationship: Postdoc Advisor
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4. **Norman Y. Yao**
   Professor, Department of Physics, Harvard University
   Relationship: Collaborator
   Email: nyao@fas.harvard.edu